Grade: Check Plus

1) ATM Probability

X	$f(x)$	μ	σ^2
10	.10	1	160
20	.15	3	135
40	.30	12	30
60	.15	9	15
80	.25	20	225
100	.05	5	125
	1.00	50	690
			26.27

a) at least \$60 $x >= 60$ 0.45 sum 45 %

✓ Qa

$.15 + .25 + .05 = .45$

✓ Qb

b) $\mu = 50$

$\sigma = \sqrt{690} = 26.27$

$\mu = x f(x)$

$\sigma^2 = (x - \mu)^2 f x$

$\sigma^2 = x^2 f(x) - \mu^2$

$\sigma^2 = \sqrt{\sigma}$

✓ Qc

c) The mean is not a single transaction. This is just the average overtime.

d) $\mu + \sigma = 50 + 26.27 = 76.27$

✓ Qd

$\mu - \sigma = 50 - 26.27 = 23.73$

$23.73 \leq X = 76.27$

$x[40, 60]$ $.3 + .15 = .45 = 45\%$

e) $\mu + 2\sigma = 102.54$

$\mu - 2\sigma = -2.54$

✓ Qe

$-2.54 \leq X \leq 102.54$

$X[10, 20, 40, 60, 80, 100] = 100\%$

All values fall within range